

VISTRA - Research Summary

Trusted-source literature, datasets, algorithm references and evidence rules for PS 26126

Evidence policy: PS-specific VISTRA content is from the supplied blueprint (B1). External technical and dataset facts are tied to the numbered source register. Project performance numbers are not claimed until measured.

1. Research scope and evidence levels

Evidence levels: B1 = supplied VISTRA blueprint; S1-S15 = public first-party or research sources; MEASURE = a project result that must come from experiment.

The blueprint contains project-specific formulas, thresholds and roadmap and should not be presented as universal standards. External papers support algorithm or dataset selection; they do not become VISTRA performance results.

Evidence: B1 Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

2. RUGD: outdoor visual perception evidence

The RUGD paper addresses unstructured outdoor environments where structured road cues are limited. It describes irregular class boundaries, diverse terrain types and difficult visual conditions such as blurred frames. The paper reports 7,453 labeled images.

This supports using RUGD as a research/validation source for outdoor terrain semantics. It does not automatically match VISTRA's six-class ontology; class mapping must be documented by the team.

Evidence: S9 RUGD paper / official host

3. RELLIS-3D: multimodal off-road evidence

The official RELLIS-3D repository reports 13,556 annotated LiDAR scans and 6,235 images, collected on the Rellis Campus of Texas A&M; University. It describes a 20-class ontology and provides RGB, stereo imagery, LiDAR, GPS and IMU data.

The dataset paper positions RELLIS-3D as a resource for off-road semantic scene understanding and notes class imbalance and environmental topography as challenges.

Evidence: **S10** RELLIS-3D official repository | **S11** RELLIS-3D paper

4. SegFormer research basis

SegFormer is presented as a semantic-segmentation framework with a hierarchical transformer encoder and a lightweight MLP decoder. The paper describes the design as efficient while retaining multi-scale features.

The Hugging Face implementation documentation lists MiT-B0 at 3.7M parameters. This supports the blueprint's small-model choice. It does not establish VISTRA FPS or latency.

Evidence: **S4** SegFormer paper | **S5** Hugging Face SegFormer documentation

5. ORB-SLAM3 research basis

ORB-SLAM3 supports visual and visual-inertial SLAM for monocular, stereo and RGB-D cameras and includes multi-map behavior. It is therefore compatible with a stereo camera-primary localization architecture.

The VISTRA project still needs its own calibration, hardware timing, failure handling and accuracy measurements.

Evidence: S6 ORB-SLAM3 paper | B1 Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

6. D* Lite research basis

The original D* Lite work describes incremental heuristic search that reuses information from previous searches and applies it to robot navigation in unknown terrain. This matches the changing VISTRA risk map and replanning need.

Evidence: S7 D* Lite - Carnegie Mellon Robotics Institute

7. DWA research basis

The Dynamic Window Approach searches reachable velocity commands and derives the search from robot motion dynamics. The original publication includes goal progress, forward velocity and obstacle distance in its objective.

VISTRA uses this general concept but adds risk-map terms and the project's safety policy.

Evidence: **S8** Dynamic Window Approach - Carnegie Mellon Robotics Institute | **B1** Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

8. Current software ecosystem

Technology	Trusted evidence
ROS 2 Jazzy	Official ROS 2 Jazzy documentation and package ecosystem
Gazebo Harmonic	Official docs; current release page lists Harmonic as LTS to May 2029
ONNX Runtime	Official docs describe a cross-platform ML inference runtime
OpenCV	Official computer-vision APIs
CVAT	Official annotation documentation
FastAPI	Official Python API framework documentation

Evidence: **S2** ROS 2 Jazzy documentation | **S3** Gazebo Harmonic documentation | **S12** ONNX Runtime documentation | **S13** CVAT documentation | **S14** OpenCV documentation | **S15** FastAPI documentation

9. Dataset comparison

Dataset	Documented fact	VISTRA use	Caution
RUGD	7,453 labeled images	Terrain semantics	Ontology mapping required
RELLIS-3D	13,556 LiDAR + 6,235 images; 20 classes	Off-road multimodal research	Different taxonomy and large files
Synthetic simulation	Controlled labels/ground truth	Closed-loop benchmark	Synthetic appearance differs from real
Own data	About 300-500 frames planned	Real-video validation	Must remain held out

Evidence: **B1** Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF) | **S9** RUGD paper / official host | **S10** RELLIS-3D official repository | **S11** RELLIS-3D paper

10. Baseline review

Dimension	Baseline	VISTRA design
Terrain representation	Binary occupancy	Continuous risk + uncertainty
Unknown hazards	Undetected can be free	Unknown/uncertain becomes costly
Global replan	A*	D* Lite
Local control	Baseline stack	DWA on risk map
Safety	Fixed-distance stop	TTC + terrain risk
Degradation	No confidence governor	NCS + Recovery Manager
Evidence	Single baseline	Baseline + fault injection + ablations

This is a project-design comparison from B1, not a measured performance ranking.

Evidence: B1 Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

11. Research gaps that require experiments

- Best real-image input resolution for the target compute device.
- Transfer quality of the six-class mapping on RUGD and RELLIS-3D.
- Frequency of stereo VO track loss on feature-poor terrain.
- Stable values for λ_u , λ_g and λ_{prox} across seeds.
- TTC thresholds that avoid both collisions and excessive nuisance stops.
- Measured contribution of each VISTRA innovation in ablation.

Evidence: B1 Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

12. Research integrity checklist

- Use primary sources for dataset statistics and current software status.
- Do not copy paper metrics into VISTRA result tables.
- Do not quote paper FPS as project FPS.
- Keep sequence/location-level train/test splits.
- Log every seed, model version and parameter set.
- Report failures and negative results.
- Keep the evidence register synchronized with the final presentation.

Evidence: **B1** Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF) | **S4** SegFormer paper | **S5** Hugging Face SegFormer documentation | **S6** ORB-SLAM3 paper | **S7** D* Lite - Carnegie Mellon Robotics Institute | **S8** Dynamic Window Approach - Carnegie Mellon Robotics Institute | **S9** RUGD paper / official host | **S10** RELLIS-3D official repository

Appendix - Trusted Source Register

B1 - Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

PS 26126 metadata, VISTRA design, formulas, roadmap, benchmark protocol, risks and limitations.

Supplied project source PDF.

S1 - Bharat Electronics Limited - Official Homepage

Official BEL organizational description.

<https://bel-india.in/homepage/>

S2 - ROS 2 Jazzy documentation

ROS 2 Jazzy documentation and packages.

<https://docs.ros.org/en/jazzy/>

S3 - Gazebo Harmonic documentation

Official Gazebo Harmonic documentation and release information.

<https://gazebosim.org/docs/harmonic/>

S4 - SegFormer paper

Original SegFormer architecture and efficiency rationale.

<https://arxiv.org/abs/2105.15203>

S5 - Hugging Face SegFormer documentation

Model configuration; MiT-B0 listed at 3.7M parameters.

https://huggingface.co/docs/transformers/main/model_doc/segformer

S6 - ORB-SLAM3 paper

Visual and visual-inertial SLAM, stereo support and multi-map behavior.

<https://arxiv.org/abs/2007.11898>

S7 - D* Lite - Carnegie Mellon Robotics Institute

Incremental heuristic replanning for unknown terrain.

<https://publications.ri.cmu.edu/d-lite>

S8 - Dynamic Window Approach - Carnegie Mellon Robotics Institute

Velocity-space local collision avoidance using robot dynamics.

<https://publications.ri.cmu.edu/the-dynamic-window-approach-to-collision-avoidance>

S9 - RUGD paper / official host

7,453 pixel-wise labeled images; unstructured outdoor driving challenges.

https://rugd.vision/pdfs/RUGD_IROS2019.pdf

S10 - RELLIS-3D official repository

13,556 LiDAR scans, 6,235 images, 20-class ontology and multimodal data.

<https://github.com/unmannedlab/RELLIS-3D>

S11 - RELLIS-3D paper

Off-road semantic scene understanding dataset motivation and benchmarks.

<https://arxiv.org/abs/2011.12954>

S12 - ONNX Runtime documentation

Cross-platform ML inference runtime.

<https://onnxruntime.ai/docs/>

S13 - CVAT documentation

Visual dataset annotation and management platform.

https://docs.cvat.ai/docs/getting_started/overview/

S14 - OpenCV documentation

Computer-vision implementation APIs.

<https://docs.opencv.org/>

S15 - FastAPI documentation

Python API framework used for dashboard integration.

<https://fastapi.tiangolo.com/>